

HW 1

Naren Manikandan

January 30, 2026

1.1

a. Binomial Distribution Formula:

$$P(x) = \binom{n}{x} p^x q^{n-x}$$

Now we can plug in our own values:

- N - number of trials
- $p = r\Delta t$ - probability of photon arriving
- k - successes desired
- $q = 1 - p$ - probability of photon not arriving

Thus, the probability of k photons arriving is:

$$\begin{aligned} P(k) &= \binom{N}{k} (r\Delta t)^k (1 - r\Delta t)^{N-k} \\ &= \binom{\frac{T}{\Delta t}}{k} (r\Delta t)^k (1 - r\Delta t)^{\frac{T}{\Delta t} - k} \end{aligned}$$

b.

Based on the given info, we'll rearrange some equations.

$$\begin{aligned} N\Delta t &= T \\ N(r\Delta t) &= rT \\ r\Delta t &= \frac{rT}{N} \end{aligned}$$

We know that rT remains constants. Let's try substituting values in $P(k)$.

$$\begin{aligned}
P(k) &= \binom{N}{k} (r\Delta t)^k (1 - r\Delta t)^{N-k} \\
&= \frac{N!}{k! (N-k)!} (r\Delta t)^k (1 - r\Delta t)^{N-k} \\
&= \frac{N!}{k! (N-k)!} \left(\frac{rT}{N}\right)^k \left(1 - \frac{rT}{N}\right)^{N-k} \\
&= \frac{N!}{k! (N-k)!} \left(\frac{rT}{N}\right)^k \left(1 - \frac{rT}{N}\right)^N \left(1 - \frac{rT}{N}\right)^{-k} \\
&= \frac{N!}{(N-k)! N^k} \frac{(rT)^k}{k!} \left(1 - \frac{rT}{N}\right)^N \left(1 - \frac{rT}{N}\right)^{-k}
\end{aligned}$$

Now let's analyze each term. As N tends to ∞ , this term becomes 1

$$\begin{aligned}
\frac{N!}{(N-k)! N^k} &= \frac{(N-1)(N-2)\dots(N-(k+1))}{N^k} \\
&= \left(1 - \frac{1}{N}\right)\left(1 - \frac{2}{N}\right)\dots \rightarrow 1
\end{aligned}$$

$\frac{(rT)^k}{k!}$ stays the same.

We know that

$$\lim_{N \rightarrow \infty} \left(1 - \frac{rT}{N}\right)^N = e^{-\frac{rT}{N}}$$

Finally,

$$\lim_{N \rightarrow \infty} \left(1 - \frac{rT}{N}\right)^{-k} = 1$$

since the exponent is constant and N is going to ∞ .

Combining these terms gives us:

$$P(k) = \frac{(rT)^k e^{-rT}}{k!}$$

c.

Let's begin by looking at the mean or expected value of the Poisson dist.

$$\begin{aligned}
P(k) &= \frac{\lambda^k e^{-\lambda}}{k!} \\
E[k] &= \sum_{k=0}^{\infty} k P(k) = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} \\
&= \lambda e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}
\end{aligned}$$

The summation term is a Taylor series! Hence:

$$\lambda e^{-\lambda} e^{-\lambda} = \lambda$$

Variance is given as:

$$Var(k) = E[k^2] - (E[k])^2$$

Let's find $E[k^2]$.

$$\begin{aligned} E[k^2] &= \sum_{k=0}^{\infty} k^2 \frac{\lambda^k e^{-\lambda}}{k!} \\ [k^2 = k(k-1) + k] \\ &= \sum_{k=0}^{\infty} k(k-1) \frac{\lambda^k e^{-\lambda}}{k!} + \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} \\ &= \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-2)!} + \lambda \\ &= e^{-\lambda} \lambda^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} + \lambda \\ &= e^{-\lambda} \lambda^2 e^{\lambda} + \lambda = \lambda^2 + \lambda \end{aligned}$$

*note: i incremented to k=1 because k=0 is 0.

We then do the final steps to get variance:

$$\lambda^2 + \lambda - (E[k])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Hence, mean and variance are both λ !!

1.2

(a)

$$\lambda = 10, CV \approx 0.315$$

$$\lambda = 100, CV \approx 0.0099$$

$$\lambda = 1000, CV \approx 0.031$$

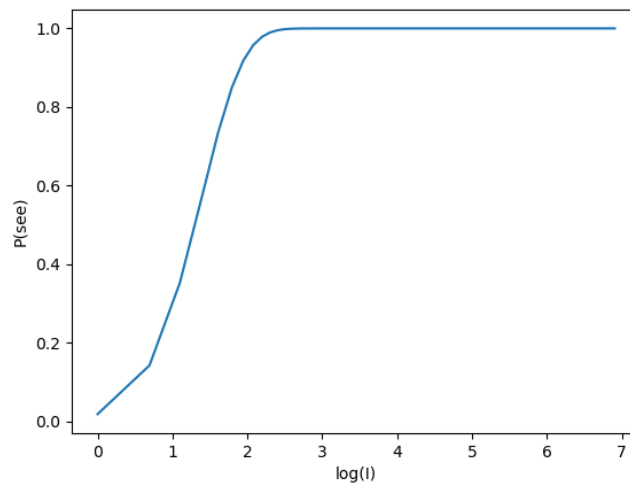
- (b) The relative noise decreases as noise intensity increases. This means that in dim conditions, the ability to measure light reliably is lower than bright conditions. This increase relationship occurs despite absolute noise increasing with the brightness of the light.

1.3

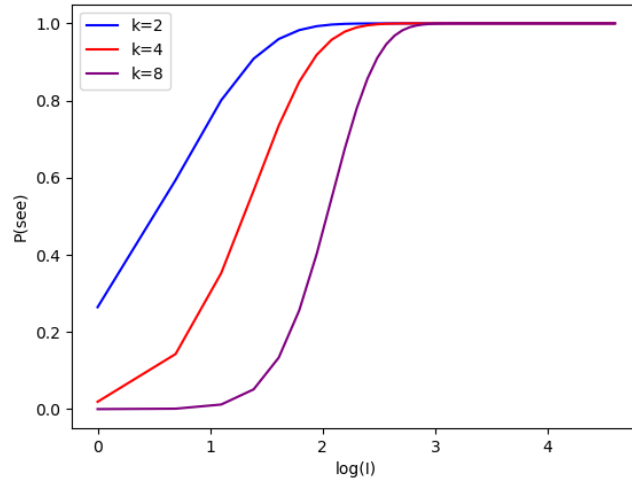
- (a) Primarily, we can see the quantization of light particles as there is some inherent variance/randomness to them (that wouldn't occur if they were waves traveling at a steady stream). Since $\text{Var} = \text{Mean}$, we can state that each photon is independent of one another. Also, this equality shows that there is always some fundamental noise in a light system no matter how precisely they're shot out, representing quantum effects.
- (b) Covariance describes how two particles move with respect to each other. It represents the correlation of the particle counts.

2.1

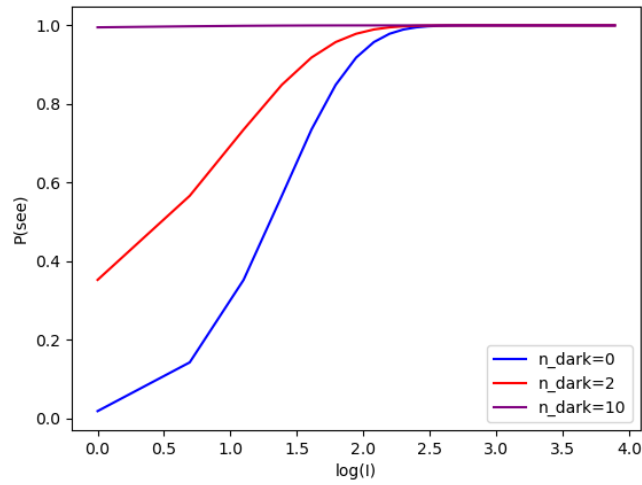
a.



b.

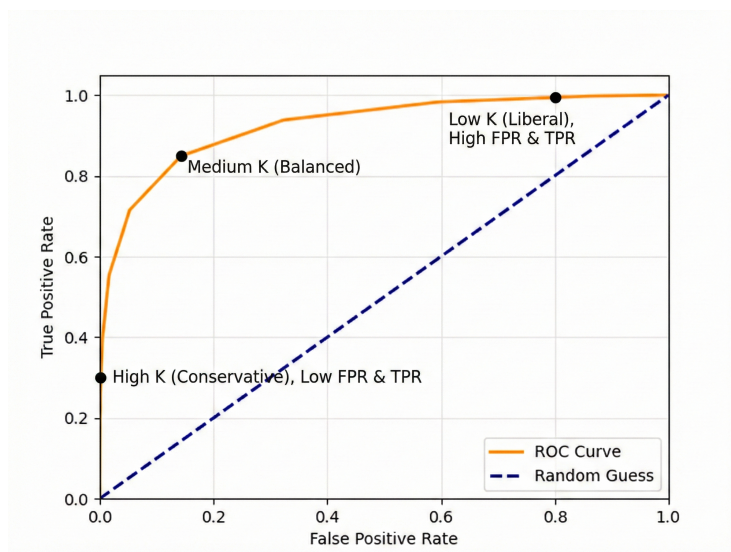


Raising the threshold lowers the left half of the curve meaning that we're getting a lower probability of seeing. This makes sense since we're bounding to higher quantities for the same response.



It looks like $n_{dark} = 0$ best fits the paper's data.

2.2

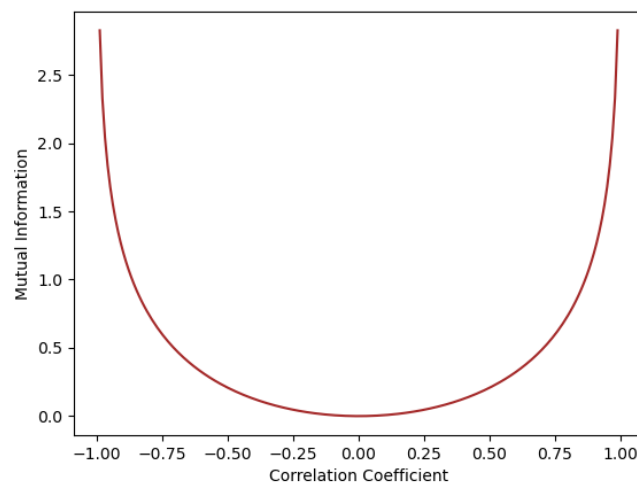


If you're being conservative and have a high k threshold, this means that you'll greatly eliminate any false positives, but also inherently limiting true positive rates since you're rejecting most of the samples.

3.1

- (a) Covariance is a statistical measure of the joint variability of two variables. Neurons driven by the same stimulus would likely have a positive covariance as they would react similarly to a certain cue.
- (b) ρ is just a normalized version of covariance which is useful when comparing against different sets of neurons since magnitude won't have an affect on this variable. It lives between -1 and 1 .

3.2



b. When $\rho \rightarrow 1$, there is a perfect correlation between the two variables and the scatter cloud tends towards a single line with positive slope. When $\rho = 1$, the scatter plot completely converges onto this line and the conditional width is 0. Hence, there's no uncertainty in the other variable if you know one which causes $I \rightarrow \infty$.

3.3

Actual MI : 0.7369655941662064
Predicted MI : 0.546323301818091

They don't match.

Bin size : 5
Actual MI : 0.7369655941662064
Predicted MI : 0.3104404420666641
Diff (P-A) : -0.42652515209954234

Bin size : 20
Actual MI : 0.7369655941662064
Predicted MI : 0.516787655192447
Diff (P-A) : -0.22017793897375948

Bin size : 100
Actual MI : 0.7369655941662064
Predicted MI : 0.6953382711065286
Diff (P-A) : -0.04162732305967787

The estimate increases with bin size since we're making more precise measurements without merging the data too much and losing information. As the bin size increases, we tend towards the actual MI given by the equation.

4.2

This homework was quite difficult and time-consuming, especially since I'm just now getting used to doing everything in latex versus handwriting everything.

The ROC trade-off clicked for me with regards to how setting k determines what true/false positive rates one can tolerate. However, I don't have a very good statistics background which makes many of these concepts hard to understand.

I do really like the graphing and analysis portions since I've never learned in this style before (from a class). I enjoy using python and experimenting with things so I'm looking forward to more of those problems in the future.